A Minimal Dynamical Model for Structural Selection: A Field Equation for the Critical Coherence Index

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Abstract

The Critical Coherence Index (CCI) has previously been introduced as a structural diagnostic that quantifies the balance between dynamical activity and coherence in nonlinear systems [1, 2]. While its empirical scaling behaviour and predictive power have been established [3, 4, 5], its dynamical origin has so far remained unclear.

In this work, we promote the CCI to a coarse-grained dynamical field $C(x,t)$ and introduce a minimal phenomenological evolution equation. The dynamics is formulated as a gradient flow of a stabilised quartic free-energy functional, which guarantees boundedness and a well-defined attractor structure. We show that the model naturally captures the emergence of ordered and disordered regimes, corresponding to low- and high-CCI states. The proposed equation provides a dynamical realisation of structural selection and establishes a direct link between coherence diagnostics, entropy–information scaling, and variational principles [6, 7]. All results are presented as a phenomenological proposal; a first-principles derivation remains an open problem.

1 Introduction

Nonlinear systems display a rich variety of behaviours, ranging from coherent structures and metastable patterns to fully chaotic regimes. The Critical Coherence Index (CCI) was recently introduced as a structural observable that captures the delicate balance between dynamical activity and coherence [1, 2].

Empirically, the CCI obeys the scaling relation

$$\mathrm{CCI} \sim \frac{\dot{S}_{cg}^{(+)}}{I_{nn}^{\alpha}}, \tag{1}$$

where $\dot{S}_{cg}^{(+)}$ is a coarse-grained activity proxy and I_{nn} denotes nearest-neighbour mutual information [3, 4, 5]. While this relation has proven to be a powerful diagnostic and predictive tool, it does not explain how coherence evolves dynamically.

The aim of this work is therefore to introduce a minimal dynamical model in which coherence itself becomes an evolving field. We interpret the scalar field $C(x,t)$ as a coarse-grained representation of the CCI and propose a simple evolution equation governing its spatiotemporal dynamics. This approach connects directly to the structural selection programme developed in the present series [6, 7]. The field $C(x,t)$ should be interpreted as an effective, coarse-grained proxy of the CCI rather than a direct microscopic observable.

2 Model Definition

We consider a scalar coherence field $C(x,t)$ obeying the nonlinear relaxation equation

$$\partial_t C = D\nabla^2 C + aC - bC^3, \tag{2}$$

where $D > 0$ is the diffusion coefficient, a controls the linear growth of coherence, and $b > 0$ provides nonlinear saturation.

This equation can be expressed as a gradient flow

$$\partial_t C = -\frac{\delta F[C]}{\delta C}, \tag{3}$$

with the free-energy functional

$$F[C] = \int \left[\frac{D}{2} |\nabla C|^2 - \frac{a}{2} C^2 + \frac{b}{4} C^4 \right] dx. \tag{4}$$

The quartic stabilisation ensures that the energy is bounded from below and that the dynamics possesses well-defined attractors, in close analogy to the structural free energy F_{struct} introduced earlier [6, 7].

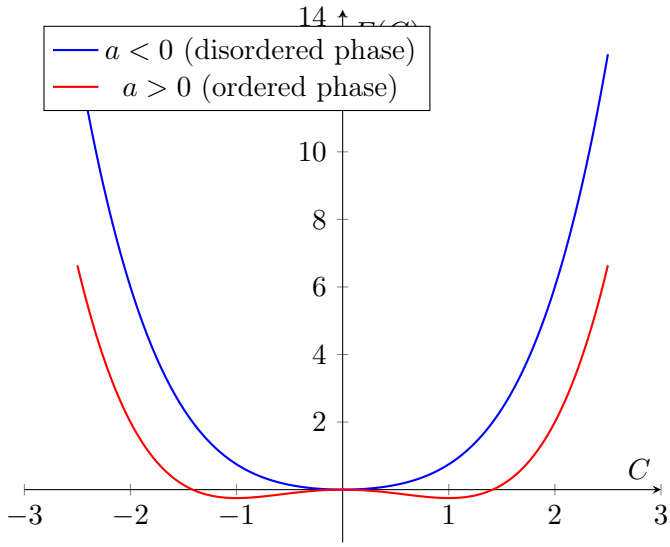


Figure 1: Free-energy landscape $F(C)$ for different values of the control parameter a. For $a < 0$ the functional has a single minimum at $C = 0$ (disordered phase). For $a > 0$ a double-well structure emerges (ordered phase).

3 Stationary Solutions and Stability

Stationary states satisfy

$$D\nabla^2 C + aC - bC^3 = 0. \tag{5}$$

In the spatially uniform case this reduces to

$$aC - bC^3 = 0 \quad \Rightarrow \quad C = 0 \quad \text{or} \quad C = \pm\sqrt{\frac{a}{b}}. \tag{6}$$

Linear stability analysis around the trivial solution $C = 0$ yields

$$\partial_t \delta C = D\nabla^2 \delta C + a\delta C. \tag{7}$$

Thus:

- For $a < 0$: $C = 0$ is stable (disordered regime).
- For $a > 0$: $C = 0$ becomes unstable and the nonzero solutions $C = \pm\sqrt{a/b}$ emerge as stable states (ordered regime).

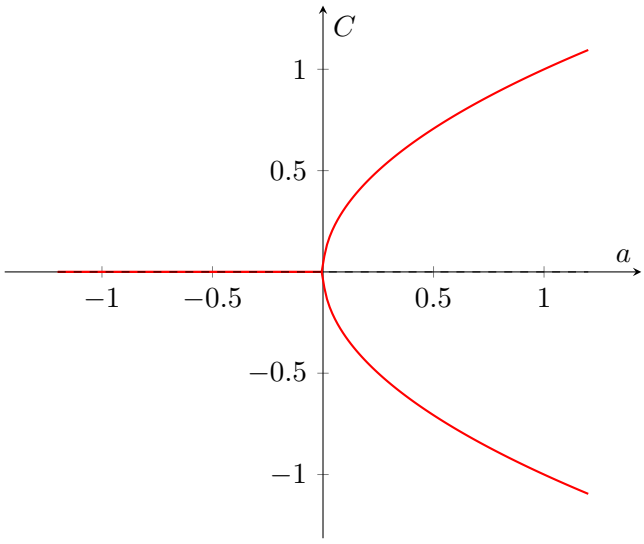


Figure 2: Bifurcation diagram of the uniform stationary solutions. For $a < 0$ only the trivial solution $C = 0$ exists. For $a > 0$ two stable nonzero branches appear, signalling a supercritical pitchfork bifurcation.

4 Numerical Illustrations

We illustrate the dynamics in both zero- and one-dimensional settings. As shown in Fig. 3, the dynamics converges toward domain-structured configurations while the free energy decreases monotonically, consistent with the gradient-flow formulation.

4.1 Zero-dimensional Dynamics

In the spatially homogeneous case the equation simplifies to

$$\partial_t C = aC - bC^3. \tag{8}$$

For parameters $a = b = 1$ the system possesses two stable fixed points at $C_* = \pm 1$. Numerical integration confirms rapid convergence to these states from generic initial conditions.

4.2 One-dimensional Field Evolution

With periodic boundary conditions and random initial data, the full equation produces a domain-structured state in which the field locally saturates at $C \approx \pm 1$. For a representative simulation we find

$$\langle C \rangle_{\text{final}} \approx -0.080, \tag{9}$$

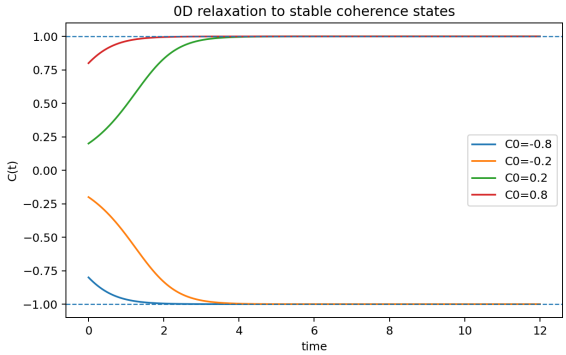
$$\text{std}(C)_{\text{final}} \approx 0.939. \tag{10}$$

The small mean value reflects the approximate balance between positive and negative domains, while the large standard deviation indicates saturation close to the stable amplitudes. The free energy decreases from $F_{\text{initial}} \approx 0.0574$ to $F_{\text{final}} \approx -27.0761$, consistent with the gradient-flow structure.

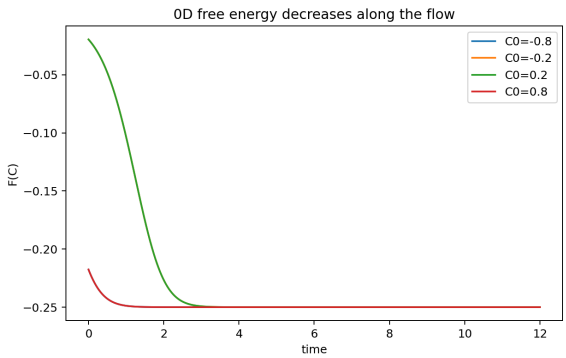
5 Interpretation

The evolution equation encodes a competition between three fundamental mechanisms:

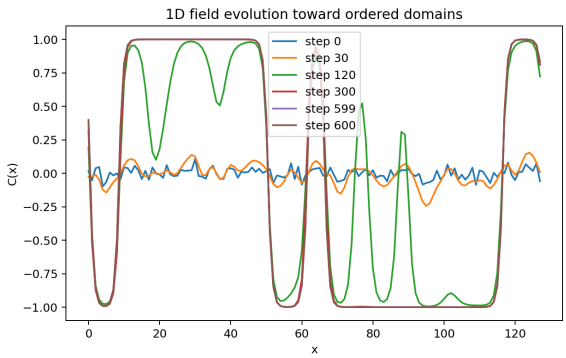
- **Diffusion** ($D\nabla^2 C$): spatial smoothing of coherence,



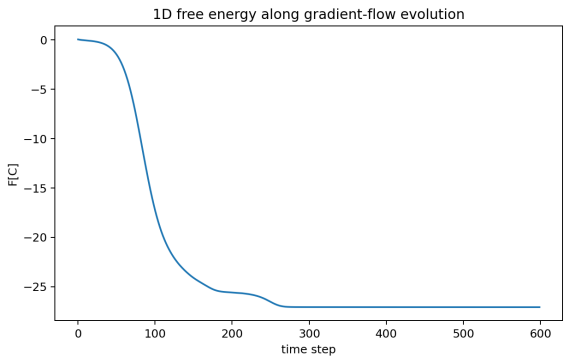
(a) Relaxation toward the stable fixed points $C = \pm\sqrt{a/b}$.



(b) Monotonic decrease of the free energy $F(C)$.



(c) 1D field evolution with domain formation.



(d) Free-energy decay in the 1D simulation.

Figure 3: Numerical solutions of the coherence-field equation (2). **Top row:** Zero-dimensional relaxation dynamics. **Bottom row:** One-dimensional evolution starting from random initial conditions, showing spontaneous domain formation and saturation near the stable values $C \approx \pm 1$. The free energy decreases monotonically in both cases.

- **Linear instability / activity** ($+aC$): growth of coherent structure,
- **Nonlinear saturation** ($-bC^3$): amplitude limitation.

These terms map directly onto the components of the CCI: dynamical activity ($\dot{S}_{cg}^{(+)}$) drives the growth of C, while structural information (I_{nn}) contributes to the stabilising nonlinear term [3, 4, 5].

6 Relation to Structural Selection

Because the dynamics is a gradient flow, the free energy satisfies

$$\frac{dF}{dt} \leq 0 \tag{11}$$

with equality only at stationary states. Structural selection thus emerges naturally as the minimisation of the phenomenological free energy $F[C]$. The stable coherent configurations correspond to the minima of this functional, providing a dynamical realisation of the Maximum-Entropy weighting on the solution manifold [6, 7].

7 Discussion

The proposed model is deliberately minimal and should be regarded as a baseline analogous to Ginzburg–Landau theory for coherence dynamics [11, 12]. It does not presuppose a microscopic definition of coherence but captures universal features of order–disorder transitions. Possible extensions include coupling to additional structural fields (in particular the five MAAT scalars), nonlocal interaction terms, embedding into quantum or statistical field theories, or explicit linkage to entropy and information observables. This suggests that the proposed dynamics captures universal features of coherence formation independent of microscopic details.

8 Conclusion

We have introduced a minimal dynamical field equation for a coarse-grained coherence field that realises the Critical Coherence Index. The model provides:

- a dynamical origin for structural selection,
- a direct connection between entropy–information scaling and field dynamics,
- a simple and transparent framework for studying coherence evolution.

This work bridges diagnostic observables and dynamical field theory, laying the foundation for more comprehensive state–structure models of complex nonlinear systems [10].

9 References

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